

Logical Bilateralism: Proofs, Models and Applications

Session 4 — λ terms and more

Sara Ayhan & Greg Restall

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λ terms

Recall unilateral natural deduction

$$\frac{\begin{array}{c} [A]^i \\ \Pi \\ B \end{array}}{A \rightarrow B} \rightarrow I^i$$

$$\frac{\begin{array}{c} \Pi \\ A \rightarrow B \end{array} \quad \begin{array}{c} \Pi' \\ A \end{array}}{B} \rightarrow E$$

Recall unilateral natural deduction

$$\begin{array}{c}
 [A]^i \\
 \Pi \\
 B \\
 \hline
 A \rightarrow B \quad \rightarrow I^i
 \end{array}
 \qquad
 \begin{array}{c}
 \Pi \qquad \Pi' \\
 A \rightarrow B \quad A \\
 \hline
 B \quad \rightarrow E
 \end{array}$$

Example Proof

$$\begin{array}{c}
 \frac{[p \rightarrow (q \rightarrow r)]^3}{q \rightarrow r} \quad \frac{[p]^1}{p} \rightarrow E \quad \frac{[p \rightarrow q]^2}{q} \quad \frac{[p]^1}{p} \rightarrow E \\
 \hline
 \frac{r}{p \rightarrow r} \rightarrow I^1 \\
 \hline
 \frac{(p \rightarrow q) \rightarrow (p \rightarrow r)}{(p \rightarrow (q \rightarrow r)) \rightarrow ((p \rightarrow q) \rightarrow (p \rightarrow r))} \rightarrow I^2 \\
 \hline
 (p \rightarrow (q \rightarrow r)) \rightarrow ((p \rightarrow q) \rightarrow (p \rightarrow r)) \rightarrow I^3
 \end{array}$$

Annotated with terms

$$\begin{array}{c}
 \frac{[x : p \rightarrow (q \rightarrow r)] \quad [z : p]}{(xz) : q \rightarrow r} \rightarrow E \quad \frac{[y : p \rightarrow q] \quad [z : p]}{(xy) : q} \rightarrow E \\
 \hline
 \frac{(xz) : q \rightarrow r \quad (xy) : q}{((xz)(xy)) : r} \rightarrow E \\
 \hline
 \frac{((xz)(xy)) : r}{\lambda z((xz)(xy)) : p \rightarrow r} \rightarrow I^z \\
 \hline
 \frac{\lambda z((xz)(xy)) : p \rightarrow r}{\lambda y \lambda z((xz)(xy)) : (p \rightarrow q) \rightarrow (p \rightarrow r)} \rightarrow I^y \\
 \hline
 \lambda x \lambda y \lambda z((xz)(xy)) : (p \rightarrow (q \rightarrow r) \rightarrow ((p \rightarrow q) \rightarrow (p \rightarrow r))) \rightarrow I^x
 \end{array}$$

Term Rules

$$\begin{array}{c}
 [x : A] \\
 \vdots \\
 M : B \\
 \hline
 x : A \quad \lambda x M : A \rightarrow B \quad \rightarrow I^x
 \end{array}
 \qquad
 \begin{array}{c}
 \vdots \qquad \qquad \vdots \\
 M : A \rightarrow B \quad N : A \\
 \hline
 (M N) : B \quad \rightarrow E
 \end{array}$$

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You can think of these *terms* as representing processes of *justification* or of *construction*.

(Justify $A \rightarrow B$ by taking A as *given*, and using this to justify B .)

You can *use* a such a justification of $A \rightarrow B$ by *applying* it to a justification of A to produce a justification of B .)

Normalisation is *evaluation* of λ terms

$$\begin{array}{c}
 [x : A] \\
 \vdots \\
 \frac{M : B}{\lambda x M : A \rightarrow B} \rightarrow I^x \quad \vdots \\
 \frac{\lambda x M : A \rightarrow B \quad N : A}{(\lambda x M N) : B} \rightarrow E
 \end{array}
 \rightsquigarrow
 \begin{array}{c}
 \vdots \\
 N : A \\
 \vdots \\
 M[x \leftarrow N] : B
 \end{array}$$

There are different ways to evaluate λ terms

- ★ CALL BY VALUE: $(\lambda x M N) \rightsquigarrow M[x \leftarrow N]$ only when N has been evaluated.
- (A λ term is a *value* if it is a variable, or a constant, or a term of the form $\lambda x M$.)

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- ★ In the CALL BY NAME discipline, we may evaluate a λ expression *before* evaluating its input.
 - In the simply typed lambda calculus, these different disciplines are different procedures for evaluating the same normal forms, as evaluation is confluent and terminating.
- ! In other contexts, as we will see, these disciplines can give different results.
e.g. to evaluate $(\lambda x(\lambda y y) D)$ by cbv where D diverges, then the result diverges.
However, a CBN evaluation path terminates: $(\lambda x(\lambda y y) D) \rightsquigarrow (\lambda y y)$.

Can we extend this to a bilateral context?

- ★ It turns out that we can, in a few different ways.

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We'll look at two different flavours:

1. A *mildly* bilateralist $\lambda\mu$ calculus due Michel Parigot (1992).

Can we extend this to a bilateral context?

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We'll look at two different flavours:

1. A *mildly* bilateralist $\lambda\mu$ calculus due Michel Parigot (1992).
2. A *thoroughly* bilateralist $\bar{\lambda}\mu\tilde{\mu}$ calculus due to Pierre-Louis Curien and Hugo Herbelin (2000).

$\lambda\mu$ calculus

Going *slightly* bilateral

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Given A and B , C follows, unless D .

Granting A and B , and setting D aside, we have C .

We can use this idea to make a *purely structural* addition
to natural deduction, keeping our operational rules unchanged.

Coordination Rules

$$\frac{\Pi \quad A \quad \cancel{A}}{\#} \uparrow$$

$$\frac{\begin{array}{c} [\cancel{A}]^i \\ \Pi \\ \# \\ A \end{array}}{\downarrow^i}$$

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$$\frac{[\cancel{A}]^i \quad \Pi \quad \#}{A} \downarrow^i$$

$$\frac{X \succ \boxed{A}, Y}{X \succ \square A, Y}$$

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$$\frac{X \succ \boxed{A}, Y}{X \succ \square A, Y}$$

$$\frac{\cancel{X} \succ \square \cancel{A}, Y}{X \succ \boxed{A}, Y}$$

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Coordination Rules

$$\frac{\Pi \quad A \quad \cancel{A} \uparrow}{\#}$$

$$\frac{[\cancel{A}]^i \quad \Pi \quad \# \quad A \downarrow^i}{A}$$

$$\frac{X \succ [A], Y}{X \succ \square A, Y}$$

$$\frac{X \succ \square A, Y}{X \succ [A], Y}$$

Here, negative polarity formulas occur only as assumptions.

An example proof

$$\begin{array}{c}
 \frac{[p]^1 \quad [\cancel{p}]^2}{\quad} \uparrow \\
 \frac{\quad}{q} \Downarrow \\
 \frac{[(p \rightarrow q) \rightarrow p]^3 \quad \frac{p \rightarrow q}{\quad} \rightarrow I^1}{p \quad} \rightarrow E \quad [\cancel{p}]^2 \uparrow \\
 \frac{\quad}{p} \Downarrow \\
 \frac{\quad}{((p \rightarrow q) \rightarrow p) \rightarrow p} \rightarrow I^3
 \end{array}$$

An example proof

$$\begin{array}{c}
 \frac{\frac{[p]^1 \quad [\cancel{p}]^2}{\quad} \uparrow}{\frac{\quad}{q} \downarrow} \# \\
 \frac{\frac{[(p \rightarrow q) \rightarrow p]^3}{p} \rightarrow E \quad \frac{\frac{p \rightarrow q}{\quad} \rightarrow I^1}{p \rightarrow q} \rightarrow E}{\quad} \rightarrow I^3 \\
 \frac{\quad}{p} \downarrow \# \\
 \frac{\quad}{((p \rightarrow q) \rightarrow p) \rightarrow p} \rightarrow I^3
 \end{array}
 \quad
 p \succ \boxed{p}$$

An example proof

$$\begin{array}{c}
 \frac{[p]^1 \quad [\cancel{p}]^2}{\quad} \uparrow \\
 \quad \quad \quad \Downarrow \quad \quad \quad \uparrow \\
 \quad \quad \quad \frac{q}{p \rightarrow q} \rightarrow I^1 \\
 \frac{[(p \rightarrow q) \rightarrow p]^3 \quad \frac{p \rightarrow q}{p \rightarrow q} \rightarrow I^1}{p} \rightarrow E \quad \quad \quad [\cancel{p}]^2 \\
 \frac{p \quad [\cancel{p}]^2}{\quad} \uparrow \\
 \quad \quad \quad \Downarrow \quad \quad \quad \uparrow \\
 \quad \quad \quad \frac{p}{p} \rightarrow I^2 \\
 \frac{\quad}{((p \rightarrow q) \rightarrow p) \rightarrow p} \rightarrow I^3
 \end{array}$$

$$p \succ \Box p$$

An example proof

$$\begin{array}{c}
 \frac{\frac{[p]^1 \quad [\cancel{p}]^2}{\quad} \uparrow}{\quad} \Downarrow \\
 \frac{\frac{[(p \rightarrow q) \rightarrow p]^3}{p} \quad \frac{\frac{p \rightarrow q}{\quad} \rightarrow I^1}{p \rightarrow q} \rightarrow E}{p} \rightarrow E \quad \frac{[\cancel{p}]^2}{\quad} \uparrow \\
 \frac{\quad}{p} \Downarrow^2 \\
 \frac{\quad}{((p \rightarrow q) \rightarrow p) \rightarrow p} \rightarrow I^3
 \end{array}
 \quad p \succ \boxed{q}, p$$

An example proof

$$\begin{array}{c}
 \frac{\frac{[p]^1 \quad [\cancel{p}]^2}{\# \downarrow} \uparrow}{q} \\
 \frac{[(p \rightarrow q) \rightarrow p]^3 \quad \frac{p \rightarrow q \rightarrow l^1}{p \rightarrow E}}{p} \rightarrow E \quad \frac{[\cancel{p}]^2}{\# \downarrow^2} \uparrow \\
 \frac{((p \rightarrow q) \rightarrow p) \rightarrow p}{\rightarrow l^3}
 \end{array}
 \quad \succ \boxed{p \rightarrow q}, p$$

An example proof

$$\begin{array}{c}
 \frac{[p]^1 \quad [\cancel{p}]^2}{\quad} \uparrow \\
 \frac{\quad}{\frac{\quad}{q} \downarrow} \Downarrow \\
 \frac{[(p \rightarrow q) \rightarrow p]^3 \quad \frac{p \rightarrow q}{\quad} \rightarrow I^1}{\quad} \rightarrow E \quad \frac{[\cancel{p}]^2}{\quad} \uparrow \\
 \frac{\quad}{\frac{\quad}{p} \downarrow^2} \Downarrow \\
 \frac{\quad}{((p \rightarrow q) \rightarrow p) \rightarrow p} \rightarrow I^3
 \end{array}
 \quad (p \rightarrow q) \rightarrow p \succ \boxed{p}, p$$

An example proof

$$\begin{array}{c}
 \frac{[p]^1 \quad [\cancel{p}]^2}{\quad} \uparrow \\
 \frac{\quad}{\frac{\quad}{q} \downarrow} \Downarrow \\
 \frac{[(p \rightarrow q) \rightarrow p]^3 \quad \frac{p \rightarrow q}{\quad} \rightarrow I^1}{p \quad} \rightarrow E \quad \frac{[\cancel{p}]^2}{\quad} \uparrow \quad (p \rightarrow q) \rightarrow p \succ \boxed{} p, p \\
 \frac{\quad}{\frac{\quad}{p} \downarrow^2} \Downarrow \\
 \frac{\quad}{((p \rightarrow q) \rightarrow p) \rightarrow p} \rightarrow I^3
 \end{array}$$

An example proof

$$\begin{array}{c}
 \frac{[p]^1 \quad [\cancel{p}]^2}{\quad} \uparrow \\
 \frac{\quad}{q} \Downarrow \\
 \frac{[(p \rightarrow q) \rightarrow p]^3 \quad \frac{p \rightarrow q}{\quad} \rightarrow I^1}{p} \rightarrow E \quad \frac{[\cancel{p}]^2}{\quad} \uparrow \\
 \frac{p}{\quad} \uparrow \\
 \frac{\quad}{p} \Downarrow^2 \\
 \frac{\quad}{((p \rightarrow q) \rightarrow p) \rightarrow p} \rightarrow I^3
 \end{array}
 \quad (p \rightarrow q) \rightarrow p \succ \boxed{p}$$

An example proof

$$\begin{array}{c}
 \frac{[p]^1 \quad [\cancel{p}]^2}{\quad} \uparrow \\
 \frac{\quad}{\frac{q}{\quad} \downarrow} \# \\
 \frac{[(p \rightarrow q) \rightarrow p]^3 \quad \frac{p \rightarrow q}{\quad} \rightarrow I^1}{p \quad \frac{\quad}{\quad} \rightarrow E} \rightarrow E \\
 \frac{\quad}{\frac{p}{\quad} \downarrow^2} \# \\
 \frac{\quad}{((p \rightarrow q) \rightarrow p) \rightarrow p} \rightarrow I^3
 \end{array}
 \succ
 \boxed{((p \rightarrow q) \rightarrow p) \rightarrow p}$$

Adding terms

$$\begin{array}{c}
 [x : A] \\
 \vdots \\
 M : B \\
 \hline
 x : A \quad \lambda x M : A \rightarrow B \quad \rightarrow^I x
 \end{array}
 \qquad
 \begin{array}{c}
 \vdots \qquad \qquad \vdots \\
 M : A \rightarrow B \quad N : A \\
 \hline
 (MN) : B \quad \rightarrow^E
 \end{array}$$

Terms

Adding terms

$$\begin{array}{c}
 [x : A] \\
 \vdots \\
 M : B \\
 \hline
 \lambda x M : A \rightarrow B \quad \rightarrow I^x
 \end{array}
 \qquad
 \begin{array}{c}
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 M : A \rightarrow B \quad N : A \\
 \hline
 (MN) : B \quad \rightarrow E
 \end{array}$$

$x : A$

$$\begin{array}{c}
 \vdots \\
 M : A \quad \alpha : \cancel{A} \\
 \hline
 \langle M | \alpha \rangle : \# \quad \uparrow
 \end{array}$$

Terms

Adding terms, labels

$$\begin{array}{c}
 [x : A] \\
 \vdots \\
 M : B \\
 \hline
 \lambda x M : A \rightarrow B \quad \rightarrow I^x
 \end{array}
 \qquad
 \begin{array}{c}
 \vdots \qquad \qquad \vdots \\
 M : A \rightarrow B \quad N : A \\
 \hline
 (MN) : B \quad \rightarrow E
 \end{array}$$

$x : A$

$$\begin{array}{c}
 \vdots \\
 M : A \quad \alpha : \cancel{A} \\
 \hline
 \langle M | \alpha \rangle : \# \quad \uparrow
 \end{array}$$

Terms

Labels

Adding terms, labels and packages

$$\begin{array}{c}
 [x : A] \\
 \vdots \\
 M : B \\
 \hline
 \lambda x M : A \rightarrow B \quad \rightarrow^I_x
 \end{array}
 \qquad
 \begin{array}{c}
 \vdots \qquad \qquad \vdots \\
 M : A \rightarrow B \quad N : A \\
 \hline
 (MN) : B \quad \rightarrow^E
 \end{array}$$

$x : A$

$$\begin{array}{c}
 \vdots \\
 M : A \quad \alpha : \cancel{A} \\
 \hline
 \langle M | \alpha \rangle : \# \quad \uparrow
 \end{array}$$

Terms

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 \vdots \qquad \qquad \vdots \\
 M : A \rightarrow B \quad N : A \\
 \hline
 (MN) : B \quad \rightarrow E
 \end{array}$$

$$\begin{array}{c}
 \vdots \\
 M : A \quad \alpha : \cancel{A} \\
 \hline
 \langle M | \alpha \rangle : \# \quad \uparrow
 \end{array}
 \qquad
 \begin{array}{c}
 [\alpha : \cancel{A}] \\
 \vdots \\
 P : \# \\
 \hline
 \mu \alpha P : A \quad \downarrow^\alpha
 \end{array}$$

Terms

Labels

Packages

The Grammar

VARIABLES

$x \quad := \quad x, y, z, \dots$

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The Grammar

VARIABLES $x \quad := \quad x, y, z, \dots$

TERMS $M, N \quad := \quad x \mid \lambda x M \mid (M N) \mid \mu \alpha P$

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These can be given a *dialogical* interpretation as assumptions, justifications, setting aside and reaching a dead end.

The Grammar

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These can be given a *dialogical* interpretation as assumptions, justifications, setting aside and reaching a dead end.

They also have a *computational* interpretation as data, constructions, continuations and processes.

A term annotated proof

$$\begin{array}{c}
 \frac{\frac{\frac{[y : p] \quad [\alpha : \cancel{p}]}{\langle y | \alpha \rangle : \#} \uparrow}{\mu\beta\langle y | \alpha \rangle : q} \downarrow^{\beta}}{\lambda y \mu\beta\langle y | \alpha \rangle : p \rightarrow q} \rightarrow^y \\
 \frac{[x : (p \rightarrow q) \rightarrow p] \quad \lambda y \mu\beta\langle y | \alpha \rangle : p \rightarrow q}{(x \lambda y \mu\beta\langle y | \alpha \rangle) : p} \rightarrow^E \\
 \frac{(x \lambda y \mu\beta\langle y | \alpha \rangle) : p \quad [\alpha : \cancel{p}]}{\langle (x \lambda y \mu\beta\langle y | \alpha \rangle) | \alpha \rangle : \#} \uparrow \\
 \frac{\langle (x \lambda y \mu\beta\langle y | \alpha \rangle) | \alpha \rangle : \#}{\mu\alpha\langle (x \lambda y \mu\beta\langle y | \alpha \rangle) | \alpha \rangle : p} \downarrow^{\alpha} \\
 \frac{\mu\alpha\langle (x \lambda y \mu\beta\langle y | \alpha \rangle) | \alpha \rangle : p}{\lambda x \mu\alpha\langle (x \lambda y \mu\beta\langle y | \alpha \rangle) | \alpha \rangle : ((p \rightarrow q) \rightarrow p) \rightarrow p} \rightarrow^x
 \end{array}$$

Negation

Given $\sharp$, it makes sense to use it to define negation:

$\neg A$ is $A \rightarrow f$, where f is the formula representative of the empty conclusion.

$$\frac{P : \sharp}{\mu P : f} fI \qquad \frac{M : f}{\langle M \rangle : \sharp} fE$$

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$$\frac{P : \sharp}{\mu P : f} fI \quad \frac{M : f}{\langle M \rangle : \sharp} fE$$

$$\frac{\frac{\frac{[x : A] \quad [\alpha : \neg A]}{\langle x | \alpha \rangle : \sharp} \uparrow}{\mu \langle x | \alpha \rangle : f} fI}{\lambda x \mu \langle x | \alpha \rangle : A \rightarrow f} \rightarrow I^x \quad \frac{y : (A \rightarrow f) \rightarrow f \quad \lambda x \mu \langle x | \alpha \rangle : A \rightarrow f}{\frac{(y \lambda x \mu \langle x | \alpha \rangle) : f}{\langle (y \lambda x \mu \langle x | \alpha \rangle) \rangle : \sharp} fE} \rightarrow E$$

$$\frac{\langle (y \lambda x \mu \langle x | \alpha \rangle) \rangle : \sharp}{\mu \alpha \langle (y \lambda x \mu \langle x | \alpha \rangle) \rangle : A} \downarrow^\alpha$$

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$$\frac{P : \sharp}{\mu P : f} fI \quad \frac{M : f}{\langle M \rangle : \sharp} fE \quad \frac{M : \neg A \quad N : A}{\langle (MN) \rangle : \sharp} \neg E \quad \frac{\begin{array}{c} [x : A] \\ \vdots \\ P : \sharp \end{array}}{\lambda x \mu P : \neg A} \neg I^x$$

$$\frac{\frac{\frac{[x : A] \quad [\alpha : \neg A]}{\langle x | \alpha \rangle : \sharp} \uparrow}{\mu \langle x | \alpha \rangle : f} fI}{\lambda x \mu \langle x | \alpha \rangle : A \rightarrow f} \rightarrow I^x \quad \frac{y : (A \rightarrow f) \rightarrow f \quad \lambda x \mu \langle x | \alpha \rangle : A \rightarrow f}{(y \lambda x \mu \langle x | \alpha \rangle) : f} \rightarrow E \quad \frac{\langle (y \lambda x \mu \langle x | \alpha \rangle) \rangle : \sharp}{\mu \alpha \langle (y \lambda x \mu \langle x | \alpha \rangle) \rangle : A} fE \downarrow \alpha$$

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$\neg A$ is $A \rightarrow f$, where f is the formula representative of the empty conclusion.

$$\begin{array}{c}
 \frac{P : \sharp}{\mu P : f} fI \quad \frac{M : f}{\langle M \rangle : \sharp} fE \quad \frac{M : \neg A \quad N : A}{\langle (MN) \rangle : \sharp} \neg E \quad \frac{[x : A] \quad \vdots \quad P : \sharp}{\lambda x \mu P : \neg A} \neg I^x
 \end{array}$$

$$\begin{array}{c}
 \frac{[x : A] \quad [\alpha : \neg A]}{\langle x | \alpha \rangle : \sharp} \uparrow \\
 \frac{\langle x | \alpha \rangle : \sharp}{\mu \langle x | \alpha \rangle : f} fI \\
 \frac{y : (A \rightarrow f) \rightarrow f \quad \lambda x \mu \langle x | \alpha \rangle : A \rightarrow f}{\langle (y \lambda x \mu \langle x | \alpha \rangle) \rangle : f} \rightarrow I^x \\
 \frac{\langle (y \lambda x \mu \langle x | \alpha \rangle) \rangle : f}{\langle \langle (y \lambda x \mu \langle x | \alpha \rangle) \rangle \rangle : \sharp} fE \\
 \frac{\langle \langle (y \lambda x \mu \langle x | \alpha \rangle) \rangle \rangle : \sharp}{\mu \alpha \langle \langle (y \lambda x \mu \langle x | \alpha \rangle) \rangle \rangle : A} \downarrow^\alpha
 \end{array}
 \quad
 \begin{array}{c}
 \frac{[x : A] \quad [\alpha : \neg A]}{\langle x | \alpha \rangle : \sharp} \uparrow \\
 \frac{\langle x | \alpha \rangle : \sharp}{\lambda x \mu \langle x | \alpha \rangle : \neg A} \neg I^x \\
 \frac{y : \neg \neg A \quad \lambda x \mu \langle x | \alpha \rangle : \neg A}{\langle \langle (y \lambda x \mu \langle x | \alpha \rangle) \rangle \rangle : \sharp} \neg E \\
 \frac{\langle \langle (y \lambda x \mu \langle x | \alpha \rangle) \rangle \rangle : \sharp}{\mu \alpha \langle \langle (y \lambda x \mu \langle x | \alpha \rangle) \rangle \rangle : A} \downarrow^\alpha
 \end{array}$$

Another proof

$$\begin{array}{c}
 \dfrac{\dfrac{\dfrac{[x : A] \quad [\alpha : \cancel{A}]}{\langle x | \alpha \rangle : \sharp} \uparrow}{\langle x | \alpha \rangle : \sharp} \neg I^x}{\dfrac{z : \neg A \rightarrow B \quad \lambda x \mu \langle x | \alpha \rangle : \neg A}{(z \lambda x \mu \langle x | \alpha \rangle) : B} \rightarrow E} \neg E \\
 \dfrac{\dfrac{[y : \neg B] \quad (z \lambda x \mu \langle x | \alpha \rangle) : B}{\langle (y (z \lambda x \mu \langle x | \alpha \rangle)) \rangle : \sharp} \neg E}{\dfrac{\mu \alpha \langle (y (z \lambda x \mu \langle x | \alpha \rangle)) \rangle : A}{\lambda y \mu \alpha \langle (y (z \lambda x \mu \langle x | \alpha \rangle)) \rangle : \neg B \rightarrow A} \downarrow^\alpha} \rightarrow I^y
 \end{array}$$

Reductions

$$\begin{array}{c}
 (\lambda x M N) \rightsquigarrow M[x \leftarrow N] \quad [x : A] \\
 \vdots \\
 \frac{M : B}{\lambda x M : A \rightarrow B} \rightarrow I^x \quad \vdots \\
 \frac{\lambda x M : A \rightarrow B \quad N : A}{(\lambda x M N) : B} \rightarrow E \rightsquigarrow \frac{N : A}{M[x \leftarrow N] : B} \rightarrow E
 \end{array}$$

Reductions

$$\begin{array}{lcl}
 (\lambda x M N) & \rightsquigarrow & M[x \leftarrow N] \\
 \langle \mu \alpha P | \beta \rangle & \rightsquigarrow & P[\alpha \leftarrow \beta]
 \end{array}
 \quad
 \begin{array}{c}
 [\alpha : \cancel{A}] \\
 \vdots \\
 P : \# \\
 \hline
 \mu \alpha P : A \quad \downarrow^{\alpha}
 \end{array}
 \quad
 \begin{array}{c}
 \beta : \cancel{A} \\
 \uparrow \\
 \langle \mu \alpha P | \beta \rangle : \#
 \end{array}
 \rightsquigarrow
 \begin{array}{c}
 [\beta : \cancel{A}] \\
 \vdots \\
 P[\alpha \leftarrow \beta] : \#
 \end{array}$$

Reductions

$$(\lambda x M N) \rightsquigarrow M[x \leftarrow N]$$

$$\langle \mu \alpha P | \beta \rangle \rightsquigarrow P[\alpha \leftarrow \beta]$$

$$\langle \mu P \rangle \rightsquigarrow P$$

$$\frac{\frac{\vdots}{P : \#} fI}{\mu P : f} fE \rightsquigarrow \frac{\vdots}{P : \#}$$

Reductions

$$(\lambda x M N) \rightsquigarrow M[x \leftarrow N]$$

$$\langle \mu \alpha P | \beta \rangle \rightsquigarrow P[\alpha \leftarrow \beta]$$

$$\langle \mu P \rangle \rightsquigarrow P$$

$$(\mu \alpha P N) \rightsquigarrow ???$$

Eliminating a retrieved formula

$$(\mu\alpha P N) \rightsquigarrow ???$$

$$\frac{\begin{array}{c} [\alpha : \cancel{A \rightarrow B}] \\ \vdots \\ P : \# \\ \hline \mu\alpha P : A \rightarrow B \end{array} \downarrow^{\alpha} \quad \begin{array}{c} \vdots \\ N : A \end{array}}{(\mu\alpha P N) : B} \rightarrow E$$

Eliminating a retrieved formula

$$(\mu\alpha P N) \rightsquigarrow ???$$

$$\begin{array}{c}
 \vdots \\
 * : A \rightarrow B \quad [\alpha : \cancel{A \rightarrow B}] \\
 \hline
 \langle * | \alpha \rangle : \sharp \quad \uparrow \\
 \vdots \\
 P : \sharp \\
 \hline
 \mu\alpha P : A \rightarrow B \quad \downarrow^{\alpha} \quad \vdots \\
 N : A \\
 \hline
 (\mu\alpha P N) : B \quad \rightarrow E
 \end{array}$$

Eliminating a retrieved formula

$$(\mu\alpha P N) \rightsquigarrow ???$$

$$\begin{array}{c}
 \vdots \\
 * : A \rightarrow B \quad [\alpha : \cancel{A \rightarrow B}] \\
 \hline
 \langle * | \alpha \rangle : \# \quad \uparrow \\
 \vdots \\
 P : \# \\
 \hline
 \mu\alpha P : A \rightarrow B \quad \downarrow^{\alpha} \quad \vdots \quad N : A \\
 \hline
 (\mu\alpha P N) : B \quad \rightarrow E
 \end{array}$$

$\rightsquigarrow$

$$\begin{array}{c}
 \vdots \quad \vdots \\
 * : A \rightarrow B \quad N : A \\
 \hline
 (* N) : B \quad \rightarrow E \quad [\beta : \cancel{B}] \quad \uparrow \\
 \hline
 \langle (* N) | \beta \rangle : \# \\
 \vdots \\
 P [\langle * | \alpha \rangle \leftarrow \langle (* N) | \beta \rangle] : \# \\
 \hline
 \mu\beta P [\langle * | \alpha \rangle \leftarrow \langle (* N) | \beta \rangle] : B \quad \downarrow^{\beta}
 \end{array}$$

Eliminating a retrieved formula

$$(\mu\alpha P N) \rightsquigarrow \mu\beta P [\langle *|\alpha \rangle \leftarrow \langle (* N)|\beta \rangle]$$

$$\begin{array}{c}
 \vdots \\
 * : A \rightarrow B \quad [\alpha : \cancel{A \rightarrow B}] \\
 \hline
 \langle *|\alpha \rangle : \# \\
 \vdots \\
 P : \# \\
 \hline
 \mu\alpha P : A \rightarrow B \quad \downarrow^\alpha \quad \vdots \\
 \hline
 (\mu\alpha P N) : B \quad \rightarrow E
 \end{array}
 \uparrow
 \rightsquigarrow
 \begin{array}{c}
 \vdots \quad \quad \quad \vdots \\
 * : A \rightarrow B \quad N : A \\
 \hline
 (* N) : B \quad \rightarrow E \quad [\beta : \cancel{B}] \\
 \hline
 \langle (* N)|\beta \rangle : \# \\
 \vdots \\
 P [\langle *|\alpha \rangle \leftarrow \langle (* N)|\beta \rangle] : \# \\
 \hline
 \mu\beta P [\langle *|\alpha \rangle \leftarrow \langle (* N)|\beta \rangle] : B \quad \downarrow^\beta
 \end{array}
 \uparrow$$

Eliminating a retrieved formula

$$\langle (\mu \alpha P N) \rangle \rightsquigarrow \mu P [\langle * | \alpha \rangle \leftarrow \langle (* N) \rangle]$$

$$\begin{array}{c}
 \vdots \\
 * : \neg A \quad [\alpha : \textcolor{red}{\neg A}] \\
 \hline
 \langle * | \alpha \rangle : \# \\
 \vdots \\
 P : \# \\
 \hline
 \mu \alpha P : \neg A \quad \downarrow \textcolor{red}{\alpha} \quad \vdots \\
 \hline
 \langle (\mu \alpha P N) \rangle : \# \quad N : A \quad \neg E
 \end{array}
 \uparrow
 \rightsquigarrow
 \begin{array}{c}
 \vdots \quad \vdots \\
 * : \neg A \quad N : A \\
 \hline
 \langle (* N) \rangle : \# \quad \neg E \\
 \vdots \\
 P [\langle * | \alpha \rangle \leftarrow \langle (* N) \rangle] : \#
 \end{array}$$

β reduction rules

$$(\lambda x M N) \rightsquigarrow M[x \leftarrow N]$$

$$\langle \mu \alpha P | \beta \rangle \rightsquigarrow P[\alpha \leftarrow \beta]$$

$$\langle \mu P \rangle \rightsquigarrow P$$

$$(\mu \alpha P N) \rightsquigarrow \mu \beta P[\langle * | \alpha \rangle \leftarrow \langle (* N) | \beta \rangle] \quad (\alpha \text{ labels a conditional})$$

$$\langle (\mu \alpha P N) \rangle \rightsquigarrow P[\langle * | \alpha \rangle \leftarrow \langle (* N) \rangle] \quad (\alpha \text{ labels a negation})$$

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$$(\mu \alpha P N) \rightsquigarrow \mu \beta P[\langle * | \alpha \rangle \leftarrow \langle (* N) | \beta \rangle] \quad (\alpha \text{ labels a conditional})$$

$$\langle (\mu \alpha P N) \rangle \rightsquigarrow P[\langle * | \alpha \rangle \leftarrow \langle (* N) \rangle] \quad (\alpha \text{ labels a negation})$$

These rules are . . . *messy*.

β reduction rules

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$$\langle (\mu \alpha P N) \rangle \rightsquigarrow P[\langle * | \alpha \rangle \leftarrow \langle (* N) \rangle] \quad (\alpha \text{ labels a negation})$$

These rules are . . . *messy*. We can simplify by going *symmetric*.

$\overline{\lambda} \mu \tilde{\mu}$ terms

Going *thoroughly* bilateral

The key bilateralist idea is that for $A, B \succ C, D$

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if C is the *conclusion* of your proof, then D is
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Going *thoroughly* bilateral

The key bilateralist idea is that for $A, B \succ \boxed{C}, D$
if C is the *conclusion* of your proof, then D is
part of the context, with *opposite polarity* to A and B . . .

$$\boxed{A}, B \succ C, D$$

. . . and we can *also* put the focus on A , and now the proof
refutes A in the context in which B is affirmed
while C, D are denied.

Going Symmetric

$$\begin{array}{c}
 \vdots \qquad \vdots \\
 M : A \quad \Phi : \cancel{A} \\
 \hline
 \langle M | \Phi \rangle : \# \quad \uparrow
 \end{array}
 \qquad
 \begin{array}{c}
 [\alpha : \cancel{A}] \\
 \vdots \\
 P : \# \\
 \hline
 \mu\alpha P : A \quad \downarrow^\alpha
 \end{array}$$

Going Symmetric

$$\begin{array}{c}
 \vdots \qquad \vdots \\
 M : A \quad \Phi : \cancel{A} \\
 \hline
 \langle M | \Phi \rangle : \# \quad \uparrow
 \end{array}
 \qquad
 \begin{array}{c}
 [\alpha : \cancel{A}] \\
 \vdots \\
 P : \# \\
 \hline
 \mu\alpha P : A \quad \downarrow^{\alpha}
 \end{array}
 \qquad
 \begin{array}{c}
 [x : A] \\
 \vdots \\
 P : \# \\
 \hline
 \tilde{\mu}x P : \cancel{A} \quad \downarrow^x
 \end{array}$$

Terms

Co-terms

Packages

Curien and Herbelin's $\rightarrow$ rules

$$\frac{[x : A] \quad \vdots \quad M : B}{\lambda x M : A \rightarrow B} \rightarrow I^x \qquad \frac{\vdots \quad M : A \quad \Phi : \cancel{B}}{(M \cdot \Phi) : \cancel{A} \rightarrow \cancel{B}} \rightarrow E$$

We can recover the *original* $\rightarrow E$ rule like this:

$$\frac{\vdots \quad M : A \rightarrow B \quad \frac{\vdots \quad N : A \quad [\alpha : \cancel{B}]}{(N \cdot \beta) : \cancel{A} \rightarrow \cancel{B}} \rightarrow E}{\langle M | (n \cdot \beta) \rangle : \sharp} \uparrow$$

$$\frac{\langle M | (n \cdot \beta) \rangle : \sharp}{\mu \alpha \langle M | (N \cdot \alpha) \rangle : B} \downarrow^\alpha$$

The Grammar

VARIABLES x $:=$ $x, y, z, \dots$

TERMS M $:=$ $x \mid \lambda x M \mid \mu \alpha P$

CO-VARIABLES α $:=$ $\alpha, \beta, \gamma, \dots$

CO-TERMS Φ $:=$ $\alpha \mid (M \cdot \Phi) \mid \tilde{\mu} x P$

PACKAGES P $:=$ $\langle M \mid \Phi \rangle$

Reduction rules

$$\frac{
 \begin{array}{c}
 [x : A] \\
 \vdots \\
 M : B
 \end{array}
 \xrightarrow{\text{I}^x}
 \frac{
 \begin{array}{c}
 \vdots \quad \vdots \\
 N : A \quad \Phi : \cancel{B}
 \end{array}
 \xrightarrow{\text{E}}
 \frac{
 \lambda x M : A \rightarrow B \quad (N \cdot \Phi) : \cancel{A \rightarrow B}
 }{
 \langle \lambda x M \mid (N \cdot \Phi) \rangle : \sharp
 }
 }{
 \langle \lambda x M \mid (N \cdot \Phi) \rangle : \sharp
 }
 \uparrow$$

Reduction rules

$$\begin{array}{c}
 [x : A] \\
 \vdots \\
 \frac{M : B}{\lambda x M : A \rightarrow B} \rightarrow I^x \quad \frac{\begin{array}{c} \vdots \\ N : A \quad \Phi : \cancel{B} \end{array}}{(N \cdot \Phi) : \cancel{A \rightarrow B}} \rightarrow E \\
 \hline
 \langle \lambda x M \mid (N \cdot \Phi) \rangle : \# \quad \uparrow
 \end{array}
 \rightsquigarrow
 \begin{array}{c}
 [x : A] \\
 \vdots \\
 \frac{M : B \quad \Phi : \cancel{B}}{\langle M \mid \Phi \rangle : \#} \uparrow \\
 \vdots \\
 \frac{N : A \quad \tilde{\mu} x \langle M \mid \Phi \rangle : \cancel{A}}{\langle N \mid \tilde{\mu} x \langle M \mid \Phi \rangle \rangle : \#} \downarrow^x \uparrow
 \end{array}$$

More reduction rules . . .

$$\frac{
 \begin{array}{c}
 [\alpha : \cancel{A}] \\
 \vdots \\
 P : \# \\
 \hline
 \mu\alpha P : A
 \end{array}
 \downarrow^{\alpha}
 \quad
 \begin{array}{c}
 \vdots \\
 \Phi : \cancel{A}
 \end{array}
 }{
 \langle \mu\alpha P \mid \Phi \rangle : \#
 }^{\uparrow}$$

More reduction rules . . .

$$\begin{array}{c}
 [\alpha : \cancel{A}] \\
 \vdots \\
 \hline
 \textcolor{green}{P} : \# \\
 \hline
 \textcolor{blue}{\mu}\alpha \textcolor{green}{P} : A \quad \downarrow^{\alpha} \quad \begin{array}{c} \vdots \\ \Phi : \cancel{A} \end{array} \\
 \hline
 \langle \textcolor{blue}{\mu}\alpha \textcolor{green}{P} \mid \Phi \rangle : \# \quad \uparrow
 \end{array}
 \rightsquigarrow
 \begin{array}{c}
 \vdots \\
 \Phi : \cancel{A} \\
 \vdots \\
 \hline
 \textcolor{green}{P} [\alpha \leftarrow \Phi] : \#
 \end{array}$$

More reduction rules . . .

$$\begin{array}{c}
 [\alpha : \cancel{A}] \\
 \vdots \\
 \frac{P : \#}{\mu\alpha P : A} \downarrow^{\alpha} \quad \begin{array}{c} \vdots \\ \Phi : \cancel{A} \end{array} \\
 \hline
 \langle \mu\alpha P \mid \Phi \rangle : \# \quad \uparrow
 \end{array}
 \rightsquigarrow
 \begin{array}{c}
 \vdots \\
 \Phi : \cancel{A} \\
 \vdots \\
 P[\alpha \leftarrow \Phi] : \#
 \end{array}$$

$$\begin{array}{c}
 [x : A] \\
 \vdots \\
 \frac{P : \#}{\tilde{\mu}x P : \cancel{A}} \downarrow^x \\
 \begin{array}{c} \vdots \\ M : A \end{array} \quad \tilde{\mu}x P : \cancel{A} \\
 \hline
 \langle M \mid \tilde{\mu}x P \rangle : \# \quad \uparrow
 \end{array}$$

More reduction rules . . .

$$\begin{array}{c}
 [\alpha : \cancel{A}] \\
 \vdots \\
 \hline
 P : \# \\
 \hline
 \mu\alpha P : A \quad \downarrow^{\alpha} \quad \begin{array}{c} \vdots \\ \Phi : \cancel{A} \end{array} \\
 \hline
 \langle \mu\alpha P \mid \Phi \rangle : \# \quad \uparrow
 \end{array}
 \rightsquigarrow
 \begin{array}{c}
 \vdots \\
 \Phi : \cancel{A} \\
 \vdots \\
 P[\alpha \leftarrow \Phi] : \#
 \end{array}$$

$$\begin{array}{c}
 [x : A] \\
 \vdots \\
 \hline
 P : \# \\
 \hline
 M : A \quad \tilde{\mu}xP : \cancel{A} \quad \downarrow^x \\
 \hline
 \langle M \mid \tilde{\mu}xP \rangle : \# \quad \uparrow
 \end{array}
 \rightsquigarrow
 \begin{array}{c}
 \vdots \\
 M : A \\
 \vdots \\
 P[x \leftarrow M] : \#
 \end{array}$$

More reduction rules . . .

$$\frac{
 \begin{array}{c}
 [\alpha : \cancel{A}] \\
 \vdots \\
 P : \# \\
 \hline
 \mu\alpha P : A
 \end{array}
 \downarrow^{\alpha}
 \begin{array}{c}
 \vdots \\
 \Phi : \cancel{A} \\
 \hline
 \uparrow
 \end{array}
 }{
 \langle \mu\alpha P \mid \Phi \rangle : \#
 }
 \rightsquigarrow
 \begin{array}{c}
 \vdots \\
 \Phi : \cancel{A} \\
 \vdots \\
 P[\alpha \leftarrow \Phi] : \#
 \end{array}$$

$$\frac{
 \begin{array}{c}
 [x : A] \\
 \vdots \\
 P : \# \\
 \hline
 \tilde{\mu}x P : \cancel{A}
 \end{array}
 \downarrow^x
 \begin{array}{c}
 \vdots \\
 M : A \\
 \hline
 \uparrow
 \end{array}
 }{
 \langle M \mid \tilde{\mu}x P \rangle : \#
 }
 \rightsquigarrow
 \begin{array}{c}
 \vdots \\
 M : A \\
 \vdots \\
 P[x \leftarrow M] : \#
 \end{array}$$

(All *substitution* is managed by these rules.)

Continuing the implication reduction

$$\begin{array}{c}
 [x : A] \\
 \vdots \\
 \frac{M : B}{\lambda x M : A \rightarrow B} \rightarrow^x \quad \frac{\begin{array}{c} \vdots \\ N : A \quad \Phi : \cancel{B} \end{array}}{(N \cdot \Phi) : \cancel{A \rightarrow B}} \rightarrow^E \quad \rightsquigarrow \quad \frac{\begin{array}{c} [x : A] \\ \vdots \\ M : B \quad \Phi : \cancel{B} \end{array}}{\langle M | \Phi \rangle : \#} \uparrow \\
 \hline
 \frac{\lambda x M : A \rightarrow B \quad (N \cdot \Phi) : \cancel{A \rightarrow B}}{\langle \lambda x M | (N \cdot \Phi) \rangle : \#} \uparrow \quad \rightsquigarrow \quad \frac{\begin{array}{c} \vdots \\ N : A \quad \tilde{\mu} x \langle M | \Phi \rangle : \cancel{A} \end{array}}{\langle N | \tilde{\mu} x \langle M | \Phi \rangle \rangle : \#} \uparrow
 \end{array}$$

Continuing the implication reduction

$$\begin{array}{c}
 [x : A] \\
 \vdots \\
 \frac{M : B}{\lambda x M : A \rightarrow B} \rightarrow |^x \quad \frac{\begin{array}{c} \vdots \quad \vdots \\ N : A \quad \Phi : \cancel{B} \end{array}}{(N \cdot \Phi) : \cancel{A \rightarrow B}} \rightarrow E \quad \rightsquigarrow \quad \frac{\begin{array}{c} [x : A] \\ \vdots \quad \vdots \\ M : B \quad \Phi : \cancel{B} \end{array}}{\langle M | \Phi \rangle : \#} \uparrow \\
 \frac{\lambda x M : A \rightarrow B \quad (N \cdot \Phi) : \cancel{A \rightarrow B}}{\langle \lambda x M | (N \cdot \Phi) \rangle : \#} \uparrow \quad \rightsquigarrow \quad \frac{\begin{array}{c} \vdots \\ N : A \end{array} \quad \frac{\langle M | \Phi \rangle : \#}{\tilde{\mu} x \langle M | \Phi \rangle : \cancel{A}} \downarrow^x}{\langle N | \tilde{\mu} x \langle M | \Phi \rangle \rangle : \#} \uparrow
 \end{array}$$

$$\rightsquigarrow \quad \frac{\begin{array}{c} \vdots \\ N : A \end{array} \quad \frac{M[x \leftarrow N] : B \quad \Phi : \cancel{B}}{\langle M[x \leftarrow N] | \Phi \rangle : \#} \uparrow$$

Continuing the implication reduction

$$\begin{array}{c}
 [x : A] \\
 \vdots \\
 \frac{M : B}{\lambda x M : A \rightarrow B} \rightarrow |^x \quad \frac{\begin{array}{c} \vdots \quad \vdots \\ N : A \quad \Phi : \cancel{B} \end{array}}{(N \cdot \Phi) : \cancel{A \rightarrow B}} \rightarrow E \quad \rightsquigarrow \quad \frac{\begin{array}{c} [x : A] \\ \vdots \quad \vdots \\ M : B \quad \Phi : \cancel{B} \end{array}}{\langle M | \Phi \rangle : \#} \uparrow \\
 \hline
 \frac{\lambda x M : A \rightarrow B \quad (N \cdot \Phi) : \cancel{A \rightarrow B}}{\langle \lambda x M | (N \cdot \Phi) \rangle : \#} \uparrow \quad \rightsquigarrow \quad \frac{\begin{array}{c} \vdots \\ N : A \end{array} \quad \frac{\langle M | \Phi \rangle : \#}{\tilde{\mu} x \langle M | \Phi \rangle : \cancel{A}} \downarrow^x}{\langle N | \tilde{\mu} x \langle M | \Phi \rangle \rangle : \#} \uparrow
 \end{array}$$

$$\rightsquigarrow \quad \frac{\begin{array}{c} \vdots \\ N : A \end{array} \quad \frac{M[x \leftarrow N] : B \quad \Phi : \cancel{B}}{\langle M[x \leftarrow N] | \Phi \rangle : \#} \uparrow$$

(So, reducing an implication performs the substitution you expect, in two steps.)

Reduction Steps

$$\langle \lambda | \cdot \rangle \quad \langle \lambda x M | (N \cdot \Phi) \rangle \rightsquigarrow \langle N | \tilde{\mu} x \langle M | \Phi \rangle \rangle$$

$$\langle \mu | \rangle \quad \langle \mu \alpha P | \Phi \rangle \rightsquigarrow P[\alpha \leftarrow \Phi]$$

$$\langle | \tilde{\mu} \rangle \quad \langle M | \tilde{\mu} x P \rangle \rightsquigarrow P[x \leftarrow M]$$

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$$\langle | \tilde{\mu} \rangle \quad \langle M | \tilde{\mu} x P \rangle \rightsquigarrow P [x \leftarrow M]$$

We have a *critical pair*. What do we do with $\langle \mu \alpha P | \tilde{\mu} x Q \rangle$?

It can reduce to $P [\alpha \leftarrow \tilde{\mu} x Q]$ and to $Q [x \leftarrow \mu \alpha P]$.

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We have a *critical pair*. What do we do with $\langle \mu \alpha P | \tilde{\mu} x Q \rangle$?

It can reduce to $P [\alpha \leftarrow \tilde{\mu} x Q]$ and to $Q [x \leftarrow \mu \alpha P]$.

These are *not* the same — if P contains no free α and Q contains no free x , they are simply P and Q respectively.

As proofs . . .

$$\begin{array}{c}
 \begin{array}{c}
 [\alpha : \cancel{A}] \\
 \vdots \\
 P : \# \\
 \hline
 \mu\alpha P : A \downarrow^{\alpha}
 \end{array}
 \quad
 \begin{array}{c}
 [x : A] \\
 \vdots \\
 Q : \# \\
 \hline
 \tilde{\mu}x Q : \cancel{A} \downarrow^x
 \end{array} \\
 \hline
 \langle \mu\alpha P \mid \tilde{\mu}x Q \rangle : \# \uparrow
 \end{array}
 \rightsquigarrow
 \begin{array}{c}
 [x : A] \\
 \vdots \\
 Q : \# \\
 \hline
 \tilde{\mu}x Q : A \downarrow^x \\
 \vdots \\
 P[\alpha \leftarrow \tilde{\mu}x Q] : \#
 \end{array}$$

$$\rightsquigarrow
 \begin{array}{c}
 [\alpha : \cancel{A}] \\
 \vdots \\
 P : \# \\
 \hline
 \mu\alpha P : A \downarrow^{\alpha} \\
 \vdots \\
 Q[x \leftarrow \mu\alpha P] : \#
 \end{array}$$

If these two proofs are identified, then *any* two proofs of $\#$ are identical!

What to do?

- Curien and Herbelin show that the two natural choices for the critical pair for $\langle \mu\alpha P \mid \tilde{\mu}x Q \rangle$ make sense in different ways.

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- Processing the $\mu\alpha$ first, to get $P[\alpha \leftarrow \tilde{\mu}x Q]$ corresponds to a *call by value* discipline.
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- Processing the $\mu\alpha$ first, to get $P[\alpha \leftarrow \tilde{\mu}x Q]$ corresponds to a *call by value* discipline.
- Processing the $\tilde{\mu}x$ first, to get $Q[x \leftarrow \mu\alpha P]$ corresponds to a *call by name* discipline.
- In the more general setting of the symmetric calculus (unlike the simply typed λ calculus), these disciplines result in different normal forms.

Simple Types and Dependent Types

- In a *simple* type theory, the class of *types* can be defined independently from the class of *terms*. In particular, the *types* cannot *include* terms.

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- In a *dependent* type theory, this restriction is lifted.
- Martin-Löf's Intuitionistic Type Theory and the Calculus of Constructions are examples of dependent type theories.
- Herbelin (2005) showed that these bilateral systems can lead to collapse when combined with simple dependent typing systems.

The grammar of our tiny dependently typed language

Formulas: $A := t = u \mid \Sigma x A$

Types: $T := A \mid \mathbb{N}$

Number variables: $x := x, y, z, \dots$

Number terms: $t, u := x \mid 0 \mid t' \mid \text{wit } p$

Proof terms: $p, q := \text{refl} \mid \text{subst } p \ q \mid (t, p) \mid \text{prf } p$

Proof terms take *formulas* as *types*. Number terms take $\mathbb{N}$ as their *type*.

Typing rules

$$\begin{array}{c}
 0 : \mathbb{N} \\
 \\
 \frac{\vdots}{t : \mathbb{N}} \\
 t' : \mathbb{N} \\
 \\
 \frac{\vdots \quad p : \Sigma x A}{\text{wit } p : \mathbb{N}}
 \end{array}$$

Typing rules

$$\begin{array}{c}
 \vdots \\
 t : \mathbb{N} \\
 \hline
 0 : \mathbb{N} \quad t' : \mathbb{N}
 \end{array}
 \quad
 \begin{array}{c}
 \vdots \\
 p : \Sigma x A \\
 \hline
 \text{wit } p : \mathbb{N}
 \end{array}$$

$$\begin{array}{c}
 \vdots \\
 p : A[x \leftarrow t] \quad t : \mathbb{N} \\
 \hline
 (t, p) : \Sigma x A
 \end{array}
 \quad
 \begin{array}{c}
 \vdots \\
 p : \Sigma x A \\
 \hline
 \text{prf } p : A[x \leftarrow \text{wit } p]
 \end{array}$$

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 \text{prf } p : A[x \leftarrow \text{wit } p]
 \end{array}$$

$$\begin{array}{c}
 t \rightsquigarrow u \\
 \hline
 \text{refl} : t = u
 \end{array}
 \quad
 \begin{array}{c}
 \vdots \quad \vdots \\
 p : t = u \quad q : B[t] \\
 \hline
 \text{subst } p \, q : B[u]
 \end{array}$$

If we add λ and μ , things get *interesting* ...

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$$\begin{array}{c}
 \frac{1 \rightsquigarrow 1}{\text{refl} : 1 = 1} \quad \frac{}{0 : \mathbb{N}} \\
 \frac{}{1 : \mathbb{N}} \\
 \hline
 (1, \text{refl}) : \Sigma x x = 1 \quad \alpha : \cancel{\Sigma x x = 1} \\
 \hline
 \langle (1, \text{refl}) | \alpha \rangle : \# \quad \uparrow \\
 \hline
 \mu\beta \langle (1, \text{refl}) | \alpha \rangle : 0 = 1 \quad \downarrow^\beta \\
 \hline
 (0, \mu\beta \langle (1, \text{refl}) | \alpha \rangle) : \Sigma x x = 1 \quad \alpha : \cancel{\Sigma x x = 1} \\
 \hline
 \langle (0, \mu\beta \langle (1, \text{refl}) | \alpha \rangle) | \alpha \rangle : \# \quad \uparrow \\
 \hline
 \mu\alpha \langle (0, \mu\beta \langle (1, \text{refl}) | \alpha \rangle) | \alpha \rangle : \Sigma x x = 1 \quad \downarrow^\alpha
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 \end{array}$$

This is not a problem, *yet*.

Let $p^!$ be $\mu\alpha \langle (0, \mu\beta \langle (1, \text{refl}) | \alpha \rangle) | \alpha \rangle$.

If we add λ and μ , things get *interesting* ...

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 \langle (0, \mu\beta \langle (1, \text{refl}) | \alpha \rangle) | \alpha \rangle : \# \quad \uparrow \\
 \hline
 \mu\alpha \langle (0, \mu\beta \langle (1, \text{refl}) | \alpha \rangle) | \alpha \rangle : \Sigma x x = 1 \quad \downarrow^\alpha
 \end{array}$$

This is not a problem, yet.

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What is wit $p^!$?

If we add λ and μ , things get *interesting* ...

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 \frac{1 \rightsquigarrow 1}{\text{refl} : 1 = 1} \quad \frac{}{0 : \mathbb{N}} \\
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 \mu\alpha \langle (0, \mu\beta \langle (1, \text{refl}) | \alpha \rangle) | \alpha \rangle : \Sigma x x = 1 \quad \downarrow^\alpha
 \end{array}$$

This is not a problem, *yet*.

Let $p^!$ be $\mu\alpha \langle (0, \mu\beta \langle (1, \text{refl}) | \alpha \rangle) | \alpha \rangle$.

What is **wit** $p^!$? To calculate this we need some reduction rules.

Natural reduction rules

$$\text{wit}(t, p) \rightsquigarrow t \qquad \text{prf}(t, p) \rightsquigarrow p \qquad \text{subst refl } p \rightsquigarrow p$$

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$$\text{wit}(t, p) \rightsquigarrow t \quad \text{prf}(t, p) \rightsquigarrow p \quad \text{subst refl } p \rightsquigarrow p$$

$$\begin{array}{c}
 [\alpha : \textcolor{red}{\Sigma x A}] \\
 \vdots \\
 P : \# \\
 \hline
 \mu\alpha P : \Sigma x A \quad \downarrow^\alpha \\
 \hline
 \text{wit } \mu\alpha P : \mathbb{N}
 \end{array}
 \rightsquigarrow
 \begin{array}{c}
 \vdots \\
 \star : \Sigma x A \\
 \hline
 \text{wit } \star : \mathbb{N} \quad [\alpha : \textcolor{red}{\mathbb{N}}] \uparrow \\
 \hline
 \langle \text{wit } \star | \alpha \rangle : \# \\
 \vdots \\
 P [\langle \star | \alpha \rangle \leftarrow \langle \text{wit } \star | \alpha \rangle] : \# \\
 \hline
 \mu\alpha P [\langle \star | \alpha \rangle \leftarrow \langle \text{wit } \star | \alpha \rangle] : \mathbb{N}
 \end{array}$$

Using these reductions:

$$\text{wit } p^! = \text{wit } \mu\alpha \langle (0, \mu\beta \langle (1, \text{refl}) | \alpha \rangle) | \alpha \rangle$$

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 &\rightsquigarrow \mu\alpha \langle \text{wit}(0, \mu\beta \langle 1 | \alpha \rangle) | \alpha \rangle
 \end{aligned}$$

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&\rightsquigarrow \mu\alpha \langle 0 | \alpha \rangle
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&\rightsquigarrow \mu\alpha \langle \text{wit}(0, \mu\beta \langle 1 | \alpha \rangle) | \alpha \rangle \\
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 \end{aligned}$$

$$\frac{\frac{0 : \mathbb{N} \quad \alpha : \cancel{\mathbb{N}}}{\langle 0 | \alpha \rangle : \#} \uparrow}{\mu\alpha \langle 0 | \alpha \rangle : \mathbb{N}} \downarrow^{\alpha} \rightsquigarrow_{\eta} 0 : \mathbb{N}$$

Using these reductions:

$$\begin{aligned}
 \text{wit } p^! &= \text{wit } \mu\alpha \langle (0, \mu\beta \langle (1, \text{refl}) | \alpha \rangle) | \alpha \rangle \\
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$$\frac{\frac{0 : \mathbb{N} \quad \alpha : \cancel{\mathbb{N}}}{\langle 0 | \alpha \rangle : \#} \uparrow}{\mu\alpha \langle 0 | \alpha \rangle : \mathbb{N}} \downarrow^{\alpha} \rightsquigarrow_{\eta} 0 : \mathbb{N}$$

So $\text{wit } p^! \rightsquigarrow 0$.

Collapse!

Collapse!

$$\begin{array}{c}
 \vdots \\
 \frac{\text{wit } p^! \rightsquigarrow 0}{\text{refl} : \text{wit } p^! = 0} \quad \frac{p^! : \Sigma x \, x = 1}{\text{prf } p^! : \text{wit } p^! = 1} \\
 \hline
 \text{subst refl } p^! : 0 = 1
 \end{array}$$

Collapse!

$$\begin{array}{c}
 \vdots \\
 \frac{\text{wit } p^! \rightsquigarrow 0}{\text{refl} : \text{wit } p^! = 0} \quad \frac{p^! : \Sigma x x = 1}{\text{prf } p^! : \text{wit } p^! = 1} \\
 \hline
 \text{subst refl } p^! : 0 = 1
 \end{array}$$

This seems . . . *not* good.

Bilateral calculi don't mix *easily* with dependent types

- Etienne Miquey (2019) in “A Classical Sequent Calculus with Dependent Types” shows how to restrict the symmetric calculus in a principled way, allowing for a rich dependent type theory without collapse.

Bilateral calculi don't mix *easily* with dependent types

- Etienne Miquey (2019) in “A Classical Sequent Calculus with Dependent Types” shows how to restrict the symmetric calculus in a principled way, allowing for a rich dependent type theory without collapse.
- Classical logic is found *inside* intuitionistic logic by way of a double negation translation, so it should not be a surprise that there is a way to combine a classical world (of *propositions*) with a wider context in which the constructive constraints apply.

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Tomorrow

Current research, future directions

Any Questions?